

Introduction

Purpose

ArchAI is an **AI-powered architectural platform** built in **Unity 6**, enabling **home design, customization, and visualization** with advanced **AI-driven planning, material selection, cost estimation, and real-time simulations**. It is designed to **outperform ArkDesign and Coohom** by integrating **intelligent space planning, code compliance checks, and seamless 3D visualization**.

Scope

The system will provide:

- ✓ A structured survey-based design input.
- ✓ AI-generated 2D architectural plans.
- ✓ Automated conversion into interactive **3D models**.
- ✓ Real-time cost estimation & material selection.
- ✓ BIM integration & compliance verification.
- ✓ VR/AR walkthroughs for immersive visualization.

Intended Users

- Homeowners planning construction or renovation.
- Architects & designers seeking AI assistance.
- Real estate developers for rapid prototyping.
- Interior designers optimizing layouts.

Technology Stack

- **Game Engine:** Unity 6 (C#)
- **AI Processing:** Python (PyTorch/TensorFlow) via Unity ML-Agents
- **3D Modeling:** Blender 4.3, Unity ProBuilder
- **UI/UX:** Unity UI Toolkit, React.js (for Web Integration)
- **Database:** Firebase/SQL
- **Cloud & APIs:** AWS, Google Maps API (for terrain mapping)

Datasets Used

- **FloorCAD dataset**
 - **Cubicasa 5k**
 - **Self-generated architectural layouts**
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System Features

User Input & Survey

Users provide design preferences via a structured survey:

- **House Type & Style** (Modern, Classic, Industrial, etc.).
- **Rooms & Spaces** (Garage, Lounge, Study, etc.).
- **Budget & Materials Preferences.**
- **Land Terrain & Climate** (for material suggestions).
- **Smart Home & Sustainability Preferences.**

Optional Custom Input

- Users can upload **sketches or written descriptions.**
- AI can **auto-generate designs** based on datasets.

2D Floor Plan Generation (AI-Driven)

- Uses **graph-based algorithms & shape grammars** for structured layouts.
- Generates **5 optimized layouts** ensuring:
 -  Structural integrity
 -  Functional zoning
 -  Budget alignment
- Style-aware **furniture & decor suggestions.**

Editing Features

- Drag-and-drop **modular editor.**
- Room & furniture placement adjustments.
- Real-time material and color modifications.

Material Selection & Cost Estimation

- **AI-recommended materials** based on:
 -  Local construction trends.
 -  House style compatibility.
 -  Cost efficiency.
- Cost breakdown:
 -  Material pricing (integrated with market data APIs).
 -  Labor costs (location-based adjustments).
 -  Construction vs. furniture budget split.

3D Model Generation & Simulation

- Converts **2D layouts into fully interactive 3D models.**
- Includes:
 -  User-selected materials & structural elements.
 -  Realistic lighting, physics, and environment mapping.

Interactive Simulation

- **First-person exploration** of the home design.
- Door interactions, lighting toggles.
- **Day/Night mode & environmental simulation.**
- VR compatibility for **full immersion.**

Finalization Options

- **3D model export** (FBX, OBJ format).
 - **2D plan download** (PDF, PNG format).
 - **Bill of Materials & Cost Breakdown.**
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3 AI-Driven Insights & Optimization

Generative Design

- Generates **multiple design variations** based on:
 -  Site dimensions
 -  User constraints
 -  Functionality & aesthetics

Site Analysis

- Uses Google Maps & terrain data for:
 -  **Sunlight exposure analysis**
 -  **Wind pattern adjustments**
 -  **Topography impact**

Performance & Structural Simulation

- Energy consumption forecasting.
- Thermal & lighting efficiency analysis.
- Structural integrity verification.

Material & Cost Optimization

- Recommends **cost-effective & sustainable materials.**
- Real-time pricing & labor rate adjustments.

BIM Integration

- Ensures **Building Information Modeling (BIM)** compatibility.
- Enables export to **Revit, ArchiCAD, and AutoCAD.**

Regulatory Compliance & Urban Planning

- **Verifies local building codes.**
- **Assesses urban impact for large-scale projects.**

Virtual Reality Visualization

- **VR-ready walkthroughs.**
- Supports Oculus, HTC Vive.

AI-Driven Design Trends

- Integrates **real-time design trends** from architecture sources & social media.
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Ethical & Practical Considerations

Human Oversight

- AI assists but does not replace architects.

Bias Prevention

- Ensures **diverse & unbiased recommendations.**

User Experience & Accessibility

- **Intuitive UI** for all skill levels.
 - Multi-device compatibility (**desktop & mobile-friendly**).
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Final Deliverables

- ✓ AI-powered **floor plan generator**
 - ✓ **Real-time cost estimation & material suggestions**
 - ✓ **3D interactive walkthroughs**
 - ✓ **BIM & Regulatory compliance**
 - ✓ **VR-enabled exploration**
 - ✓ Fully integrated **Unity-based** architectural AI system
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